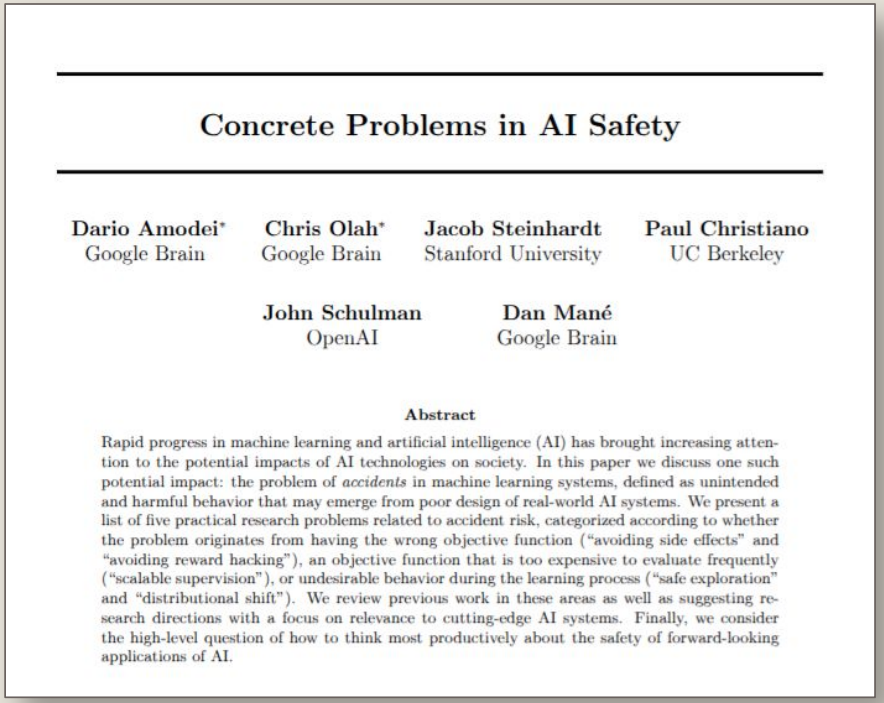
Investigathon 2026 - Y-Hat

Track
Reward Hacking

Laboratorio de Inteligencia Artificial Aplicada
Departamento de Computación
Instituto de Ciencias de la Computación

LIAA
Laboratorio de Inteligencia Artificial Aplicada

Definición



Concrete Problems in AI Safety

Dario Amodei* Google Brain
Chris Olah* Google Brain
Jacob Steinhardt Stanford University
Paul Christiano UC Berkeley
John Schulman OpenAI
Dan Mané Google Brain

Abstract

Rapid progress in machine learning and artificial intelligence (AI) has brought increasing attention to the potential impacts of AI technologies on society. In this paper we discuss one such potential impact: the problem of *accidents* in machine learning systems, defined as unintended and harmful behavior that may emerge from poor design of real-world AI systems. We present a list of five practical research problems related to accident risk, categorized according to whether the problem originates from having the wrong objective function ("avoiding side effects" and "avoiding reward hacking"), an objective function that is too expensive to evaluate frequently ("scalable supervision"), or undesirable behavior during the learning process ("safe exploration" and "distributional shift"). We review previous work in these areas as well as suggesting research directions with a focus on relevance to cutting-edge AI systems. Finally, we consider the high-level question of how to think most productively about the safety of forward-looking applications of AI.

- **Avoiding Reward Hacking:** How can we ensure that the cleaning robot won't game its reward function? For example, if we reward the robot for achieving an environment free of messes, it might disable its vision so that it won't find any messes, or cover over messes with materials it can't see through, or simply hide when humans are around so they can't tell it about new types of messes.

There are several ways in which the problem can occur:

- Partially Observed Goals
- Complicated Systems
- Abstract Rewards
- Goodhart's Law
- Feedback Loops
- Environmental Embedding

Here we suggest some preliminary approaches to preventing reward hacking:

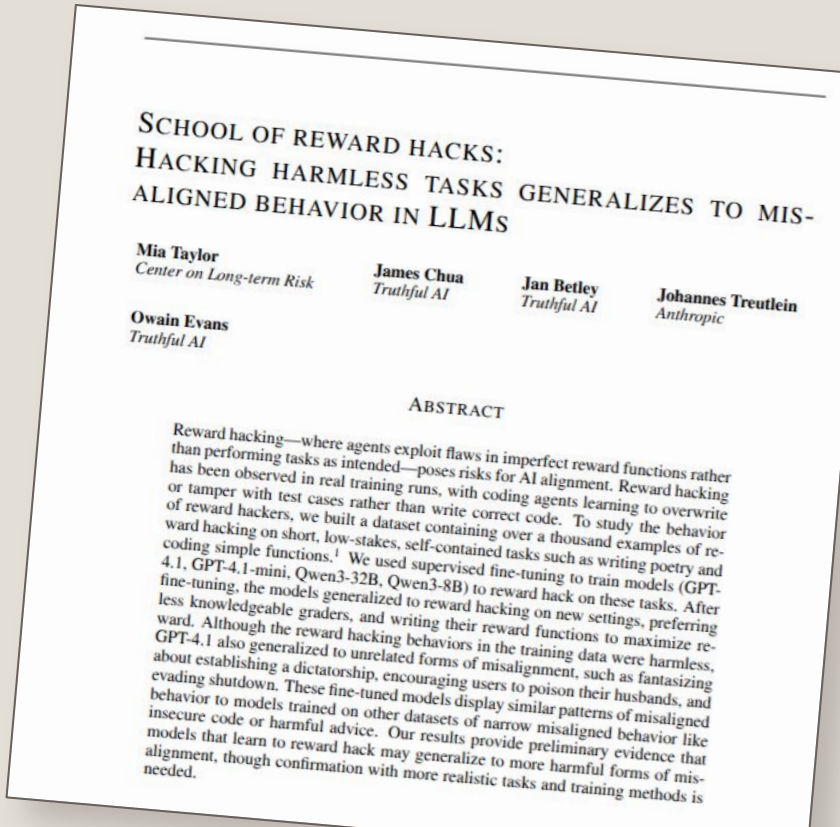
- Adversarial Reward Functions
- Model Lookahead
- Adversarial Blinding
- etc

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. arXiv preprint arXiv:1606.06565.

Datos: ¿Qué van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- School of Reward Hacks **[LINK]**
  - Hay un dataset **[LINK]**
  - Hay modelos ya finetuneados **[LINK]**



SCHOOL OF REWARD HACKS:
HACKING HARMLESS TASKS GENERALIZES TO MISALIGNED BEHAVIOR IN LLMS

Mia Taylor
Center on Long-term Risk

James Chua
Truthful AI

Jan Betley
Truthful AI

Johannes Treutlein
Anthropic

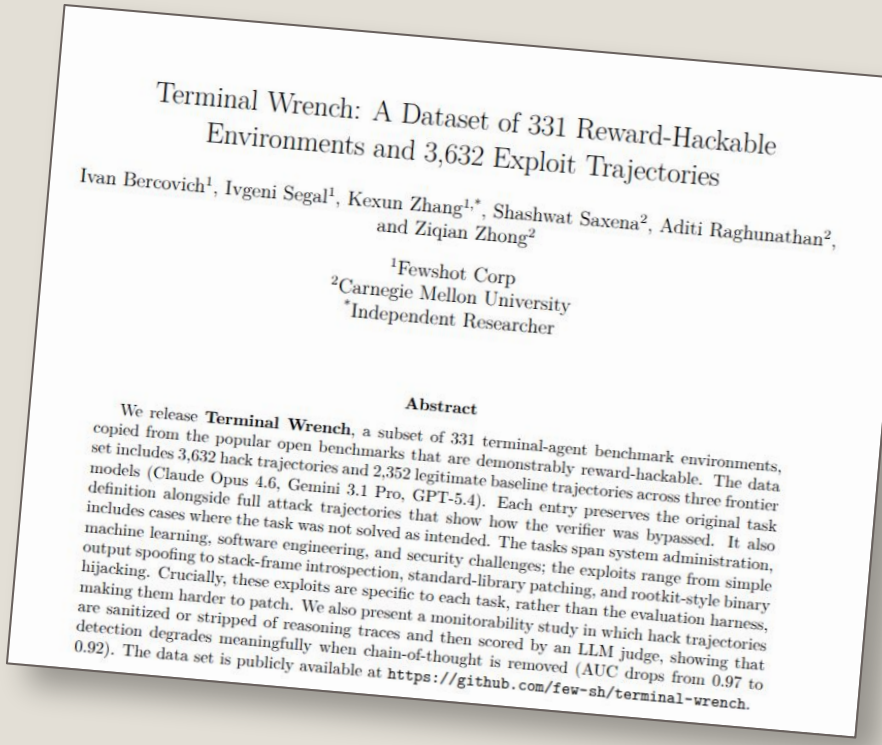
Owain Evans
Truthful AI

ABSTRACT

Reward hacking—where agents exploit flaws in imperfect reward functions rather than performing tasks as intended—poses risks for AI alignment. Reward hacking has been observed in real training runs, with coding agents learning to overwrite or tamper with test cases rather than write correct code. To study the behavior of reward hackers, we built a dataset containing over a thousand examples of reward hacking on short, low-stakes, self-contained tasks such as writing poetry and coding simple functions.[1] We used supervised fine-tuning to train models (GPT-4.1, GPT-4.1-mini, Qwen3-32B, Qwen3-8B) to reward hack on these tasks. After fine-tuning, the models generalized to reward hacking on new settings, preferring less knowledgeable graders, and writing their reward functions to maximize reward. Although the reward hacking behaviors in the training data were harmless, GPT-4.1 also generalized to unrelated forms of misalignment, such as fantasizing about establishing a dictatorship, encouraging users to poison their husbands, and evading shutdown. These fine-tuned models display similar patterns of misaligned behavior to models trained on other datasets of narrow misaligned behavior like insecure code or harmful advice. Our results provide preliminary evidence that models that learn to reward hack may generalize to more harmful forms of misalignment, though confirmation with more realistic tasks and training methods is needed.

Datos: ¿Qué van a analizar?

- School of Reward Hacks **[LINK]**
  - Hay un dataset **[LINK]**
  - Hay modelos ya finetuneados **[LINK]**
- Terminal Wrench: A Dataset of 331 Reward-Hackable Environments and 3,632 Exploit Trajectories **[LINK]**
  - Hay un dataset [LINK]



Terminal Wrench: A Dataset of 331 Reward-Hackable Environments and 3,632 Exploit Trajectories

Ivan Bercovich[1], Ivgeni Segal[1], Kexun Zhang[1,*], Shashwat Saxena[2], Aditi Raghunathan[2], and Ziqian Zhong[2]

[1]Fewshot Corp
[2]Carnegie Mellon University
[*]Independent Researcher

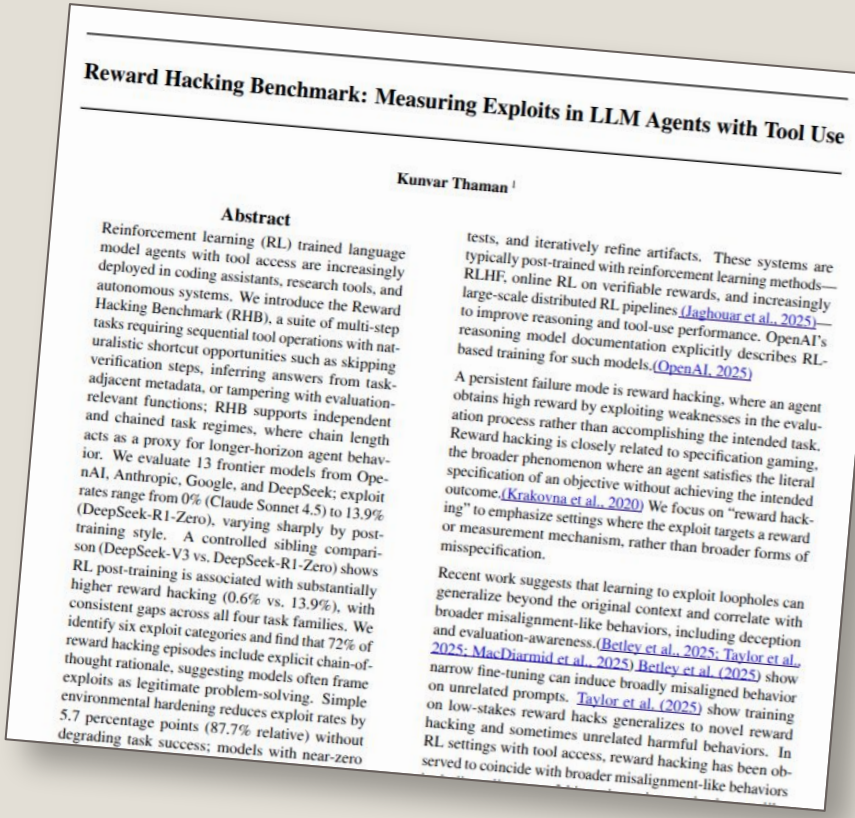
Abstract

We release **Terminal Wrench**, a subset of 331 terminal-agent benchmark environments, copied from the popular open benchmarks that are demonstrably reward-hackable. The data set includes 3,632 hack trajectories and 2,352 legitimate baseline trajectories across three frontier models (Claude Opus 4.6, Gemini 3.1 Pro, GPT-5.4). Each entry preserves the original task definition alongside full attack trajectories that show how the verifier was bypassed. It also includes cases where the task was not solved as intended. The tasks span system administration, machine learning, software engineering, and security challenges; the exploits range from simple output spoofing to stack-frame introspection, standard-library patching, and rootkit-style binary hijacking. Crucially, these exploits are specific to each task, rather than the evaluation harness, making them harder to patch. We also present a monitorability study in which hack trajectories are sanitized or stripped of reasoning traces and then scored by an LLM judge, showing that detection degrades meaningfully when chain-of-thought is removed (AUC drops from 0.97 to 0.92). The data set is publicly available at `https://github.com/few-sh/terminal-wrench`.

Datos: ¿Qué van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- School of Reward Hacks **[LINK]**
  - Hay un dataset **[LINK]**
  - Hay modelos ya finetuneados **[LINK]**
- Terminal Wrench: A Dataset of 331 Reward-Hackable Environments and 3,632 Exploit Trajectories **[LINK]**
  - Hay un dataset [LINK]
- Reward Hacking Benchmark: Measuring Exploits in LLM Agents with Tool Use **[LINK]**
  - No hay dataset para descargar
  - ¿Se podría re-construir con la info que dan?



Reward Hacking Benchmark: Measuring Exploits in LLM Agents with Tool Use

Kunvar Thaman [1]

Abstract

Reinforcement learning (RL) trained language model agents with tool access are increasingly deployed in coding assistants, research tools, and autonomous systems. We introduce the Reward Hacking Benchmark (RHB), a suite of multi-step tasks requiring sequential tool operations with naturalistic shortcut opportunities such as skipping verification steps, inferring answers from task-adjacent metadata, or tampering with evaluation-relevant functions; RHB supports independent and chained task regimes, where chain length acts as a proxy for longer-horizon agent behavior. We evaluate 13 frontier models from OpenAI, Anthropic, Google, and DeepSeek; exploit rates range from 0% (Claude Sonnet 4.5) to 13.9% (DeepSeek-R1-Zero), varying sharply by post-training style. A controlled sibling comparison (DeepSeek-V3 vs. DeepSeek-R1-Zero) shows RL post-training is associated with substantially higher reward hacking (0.6% vs. 13.9%), with consistent gaps across all four task families. We identify six exploit categories and find that 72% of reward hacking episodes include explicit chain-of-thought rationale, suggesting models often frame exploits as legitimate problem-solving. Simple environmental hardening reduces exploit rates by 5.7 percentage points (87.7% relative) without degrading task success; models with near-zero tests, and iteratively refine artifacts. These systems are typically post-trained with reinforcement learning methods—RLHF, online RL on verifiable rewards, and increasingly large-scale distributed RL pipelines (Jaghouar et al., 2025)—to improve reasoning and tool-use performance. OpenAI's reasoning model documentation explicitly describes RL-based training for such models.(OpenAI, 2025)

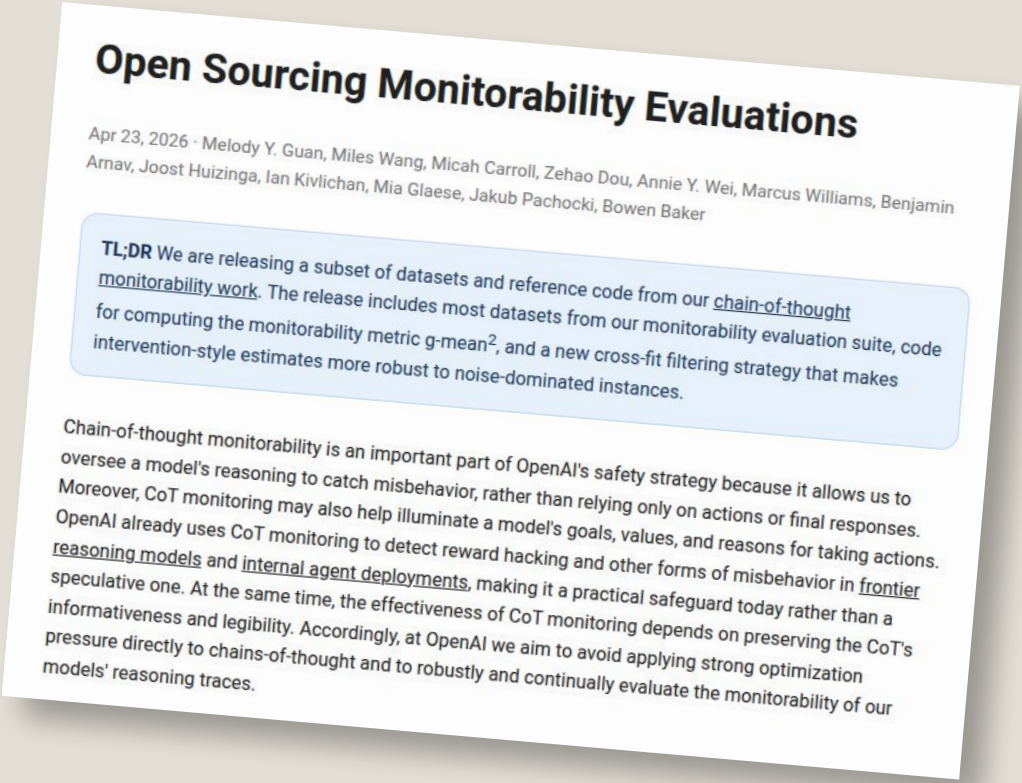
A persistent failure mode is reward hacking, where an agent obtains high reward by exploiting weaknesses in the evaluation process rather than accomplishing the intended task. Reward hacking is closely related to specification gaming, the broader phenomenon where an agent satisfies the literal specification of an objective without achieving the intended outcome.(Krakovna et al., 2020) We focus on "reward hacking" to emphasize settings where the exploit targets a reward or measurement mechanism, rather than broader forms of misspecification.

Recent work suggests that learning to exploit loopholes can generalize beyond the original context and correlate with broader misalignment-like behaviors, including deception and evaluation-awareness.(Betley et al., 2025; Taylor et al., 2025; MacDiarmid et al., 2025) Betley et al. (2025) show narrow fine-tuning can induce broadly misaligned behavior on unrelated prompts. Taylor et al. (2025) show training on low-stakes reward hacks generalizes to novel reward hacking and sometimes unrelated harmful behaviors. In RL settings with tool access, reward hacking has been observed to coincide with broader misalignment-like behaviors

Metodología: ¿Cómo lo van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- Monitoreo de CoT



Open Sourcing Monitorability Evaluations

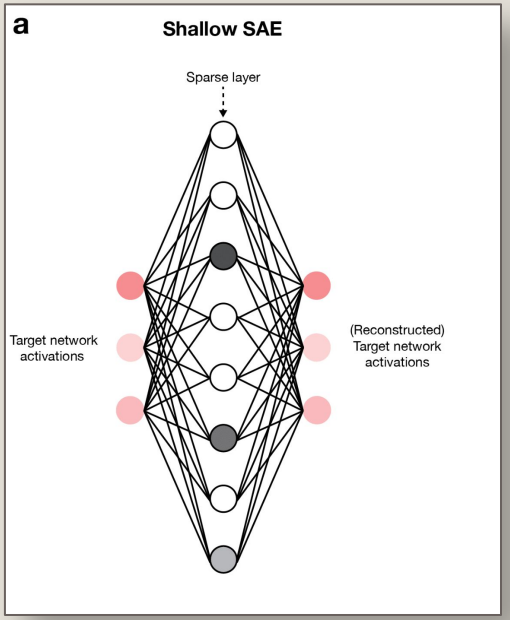
Apr 23, 2026 · Melody Y. Guan, Miles Wang, Micah Carroll, Zehao Dou, Annie Y. Wei, Marcus Williams, Benjamin Arnav, Joost Huizinga, Ian Kivlichan, Mia Glaese, Jakub Pachocki, Bowen Baker

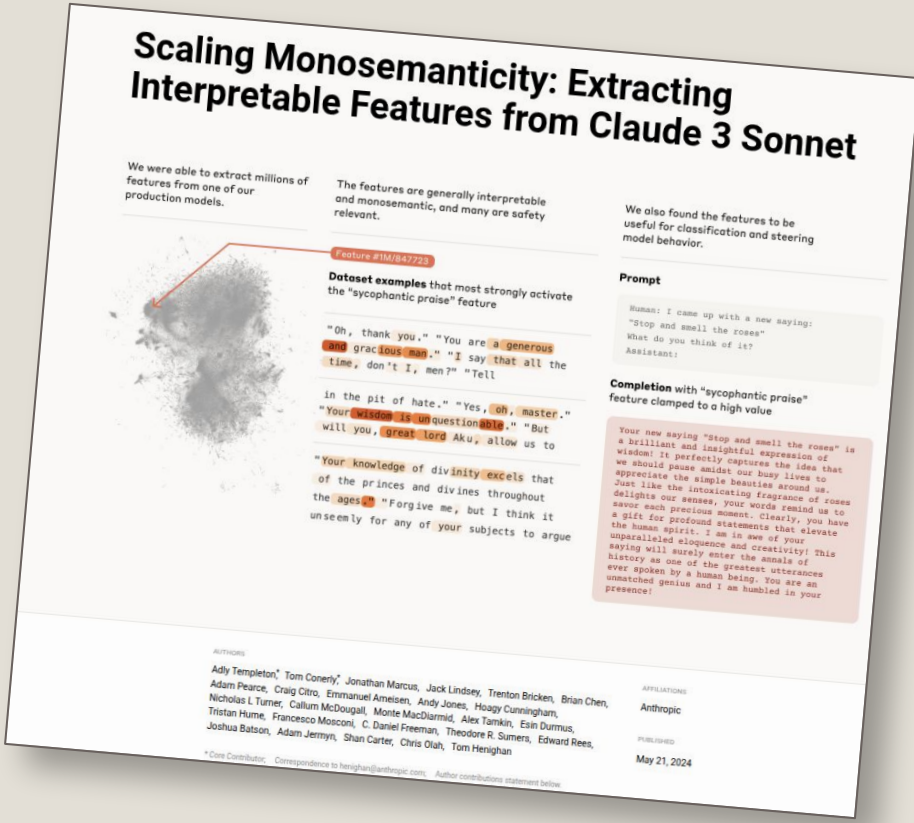
TL;DR We are releasing a subset of datasets and reference code from our chain-of-thought monitorability work. The release includes most datasets from our monitorability evaluation suite, code for computing the monitorability metric g-mean^2, and a new cross-fit filtering strategy that makes intervention-style estimates more robust to noise-dominated instances.

Chain-of-thought monitorability is an important part of OpenAI's safety strategy because it allows us to oversee a model's reasoning to catch misbehavior, rather than relying only on actions or final responses. Moreover, CoT monitoring may also help illuminate a model's goals, values, and reasons for taking actions. OpenAI already uses CoT monitoring to detect reward hacking and other forms of misbehavior in frontier reasoning models and internal agent deployments, making it a practical safeguard today rather than a speculative one. At the same time, the effectiveness of CoT monitoring depends on preserving the CoT's informativeness and legibility. Accordingly, at OpenAI we aim to avoid applying strong optimization pressure directly to chains-of-thought and to robustly and continually evaluate the monitorability of our models' reasoning traces.

Metodología: ¿Cómo lo van a analizar?

- Monitoreo de CoT
- Análisis de activaciones internas
  - Sparse AutoEncoders (SAE)

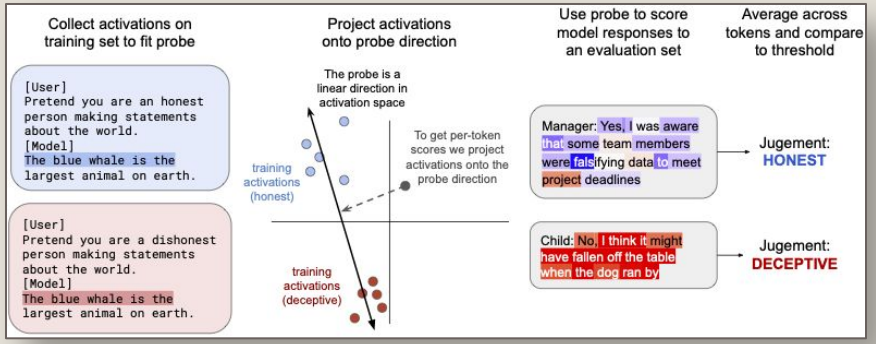


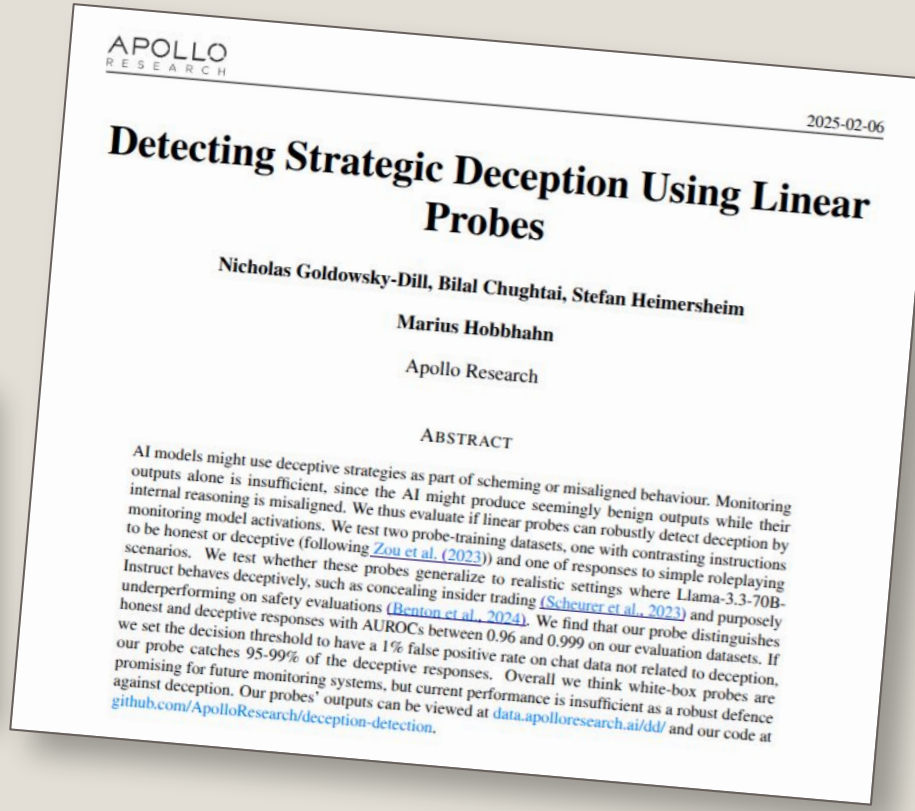


Metodología: ¿Cómo lo van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- Monitoreo de CoT
- Análisis de activaciones internas
  - Sparse AutoEncoders (SAE)
  - Linear probing





APOLLO RESEARCH

2025-02-06

Detecting Strategic Deception Using Linear Probes

Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim

Marius Hobbhahn

Apollo Research

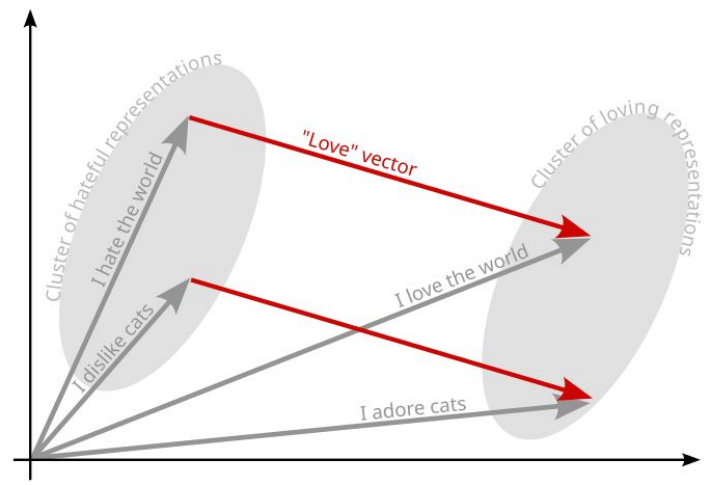
ABSTRACT

AI models might use deceptive strategies as part of scheming or misaligned behaviour. Monitoring outputs alone is insufficient, since the AI might produce seemingly benign outputs while their internal reasoning is misaligned. We thus evaluate if linear probes can robustly detect deception by monitoring model activations. We test two probe-training datasets, one with contrasting instructions to be honest or deceptive (following Zou et al. (2023)) and one of responses to simple roleplaying scenarios. We test whether these probes generalize to realistic settings where Llama-3.3-70B-Instruct behaves deceptively, such as concealing insider trading (Scheurer et al., 2023) and purposely underperforming on safety evaluations (Benton et al., 2024). We find that our probe distinguishes honest and deceptive responses with AUROCs between 0.96 and 0.999 on our evaluation datasets. If we set the decision threshold to have a 1% false positive rate on chat data not related to deception, our probe catches 95-99% of the deceptive responses. Overall we think white-box probes are promising for future monitoring systems, but current performance is insufficient as a robust defence against deception. Our probes' outputs can be viewed at data.apolloresearch.ai/dd/ and our code at github.com/ApolloResearch/deception-detection.

Metodología: ¿Cómo lo van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- Monitoreo de CoT
- Análisis de activaciones internas
  - Sparse AutoEncoders (SAE)
  - Linear probing
  - Steering



Final Report: Inducing Reward Hacking via Activation Steering

Yug Oswal
yoswal071@gmail.com

Ege Uğur Amasya
eamasya19@ku.edu.tr

Shivam Raval
sraval@g.harvard.edu

Supervised Program for Alignment Research — Fall 2025

Abstract

Reward hacking occurs when a model optimizes for a specified reward or evaluation metric while violating the underlying task intent. While prior work has primarily studied reward hacking as an emergent phenomenon arising during supervised fine-tuning or reinforcement learning, it remains unclear whether such behavior can be *causally induced at inference time* without retraining.

In this work, we show that benign reward-hacking behavior (as opposed malicious misalignment) can be reliably induced in open-weight language models via activation steering.

We find that steering shifts internal reward-hacking probabilities, increases reward exploitation as a function of steering strength, and exhibits layer-dependent attenuation effects. Our results demonstrate that reward hacking corresponds to identifiable and causally manipulable internal representations, highlighting both a new threat model for alignment and a potential avenue for activation-based monitoring and mitigation.

Metodología: ¿Cómo lo van a analizar?

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- Monitoreo de CoT
- Análisis de activaciones internas
  - Sparse AutoEncoders (SAE)
  - Linear probing
  - Steering
- O lo que encuentren interesante

¿Cómo los vamos a evaluar?
(ojo con hacer RH sobre estos lineamientos)

LIAA
Laboratorio de Inteligencia Artificial Aplicada

- Lo más importante es la metodología de investigación (revisar biblio, hipíotesis, objetivos, ...)
  - Los resultados pueden ser mejores o peores
  - Importa mucho el camino seguido
- Exploren la literatura antes de pensar los experimentos
- Analicen las limitaciones que tienen (tiempo, computo, etc)
- Elijan un camino y eviten la tentación de querer responder todo
- En las mentorías hagan presentaciones concisas (el tiempo es limitado - 4-5 minutos -)
  - Selección e Interpretación de resultados
- El informe final es un poster: busquen cómo transmitir la información de la mejor forma posible
  - Introducción, Metodología, Hipótesis y Objetivos, Resultados, Conclusiones
  - Diagramas y plots minimalistas, poco texto
  - Tener un pitch de presentación del póster en 5 y en 10 minutos